

Maaz Qureshi

Research Engineer (ML, Embodied AI and Robotics): MASc Mechtronics University of Waterloo

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | +1 437 870 8791 | m23qures@uwaterloo.ca | Waterloo, Canada

SKILLS

Machine Learning & Foundation Models: Vision-Language-Action models (π_0 , $\pi_{0.5}$), World-Action Models (literature), Parameter-Efficient Fine-Tuning (PEFT: LoRA), CNNs, VLMs, Quantization (FP16/INT8), Reinforcement Learning (PPO, policy optimization), Imitation Learning (behavior cloning) & Diffusion Policy

Robotics: 6D Pose Estimation, Visual-Inertial SLAM (volumetric mapping & localization), Motion Planning for 7-DoF Manipulators (IK/FK), Sensor Fusion (stereo RGB-D, LiDAR, mmWave 4D radar, IMU), 5G-Connected Factory Robotics, AR/XR (ARKit)

Languages: Python, C++, PyTorch, TensorRT (CUDA), Eigen, NumPy, ROS2

Tools: Docker, CI/CD, gRPC, NVIDIA Isaac, MuJoCo, Gazebo, RViz, Asynchronous GPU Pipeline, Debugging/Monitoring (Valgrind, GDB, Jtop)

RESEARCH EXPERIENCE

Embodied AI Robotics RA (Research Assistant)

May 2026 – Present

Apple Inc. (joint research with University of Waterloo)

Waterloo, Canada

- Confidential NDA project with Apple Robotics and University of Waterloo for 7-DoF manipulator motion planning.
- Built an iPhone Pro (ARKit)-based egocentric human-demonstration data pipeline for PEFT (LoRA-based) fine-tuning of foundation models.
- Fine-tuning the π -family model for Apple factory tasks on the Franka Panda arm; a world-action model pipeline is in progress (literature).
- Characterized ARKit pose-tracking accuracy against Franka Panda 6D pose and gripper-width ground truth across a 5–65°/s angular-velocity sweep, identifying a sub-40°/s operating envelope (mean translational error <10%) for reliable demonstration collection.
- Developing a UMI-style handheld gripper with a custom Franka-flange mount for scalable demonstration collection and deployment.

Robotics Software Engineer [\[video\]](#)

September 2025 – April 2026

Cobionix Corporation Inc.

Kitchener, Canada

- Developed a 22 FPS real-time multimodal RGB-D ML pipeline (SmolVLM, pose estimation, object detection, Segmentation) in C++/CUDA.
- Applied PEFT (LoRA) to fine-tune the SmolVLM vision-language backbone for object recognition.
- Applied TensorRT FP16/INT8 quantization in async GPU pipelines on NVIDIA Jetson to boost throughput and cut latency.
- Implemented real-time RGB-D stereo sensor fusion with intrinsic/extrinsic spatiotemporal calibration (>90% alignment accuracy).

Graduate Research Student: ML Robot Perception & Manipulation

January 2024 – August 2025

University of Waterloo, RoboHub Lab [\[Publication\]](#), [Video](#), [GitHub](#)

Waterloo, Canada

- Developed a distributed multi-robot dense volumetric SLAM system (C++/ROS2 Humble) with TSDF-octree mapping and ORB-SLAM3.
- Achieved an 85% improvement in map coverage over the state-of-the-art baseline across 4 autonomous mobile robots (AMRs).
- Fused mmWave 4D radar with RGB-D sensing for perception in cluttered, low-visibility settings (92% detection accuracy).

MITACS Industrial Research Internships:

Keysight Technologies Inc. [\[Publication\]](#), [Video](#), [GitHub](#)

Thesis Internship

- (SIM-to-REAL) Developed an OMPL/RRT* sampling based motion-planning pipeline for 7-DoF Franka and KUKA collaborative robots.
- Benchmarked a PPO reinforcement-learning planner against RRT sampling-based motion planning for manipulator trajectories.
- Validated sim-to-real deployment, achieving 100% joint-limit and collision avoidance across hemispherical scan patterns.
- Built with Python, ROS/RViz, and Docker for end-to-end pipeline development and deployment.

Rogers Communications Inc.

Thesis Internship

- Integrated 5G connectivity into a distributed software pipeline on NVIDIA GPUs for smart-factory robots, achieving a 37% latency reduction.
- Benchmarked 5G sub-6 GHz (3.5 GHz) and mmWave (28 GHz) channels against WiFi for industrial robotics applications.

Asset Integrity Engineer: (Robotics & ML)

May 2022 – December 2023

ABYSS Solutions Inc.

Islamabad, Pakistan (Onsite) & Remote Australia

- Performed anomaly outlier detection (supervised ML, U-Net segmentation) for corrosion detection on oil/gas platforms.
- Performed offline SLAM mapping visual model for mooring-chain and dam inspection using underwater ROV imagery.[\[Link\]](#)
- Led a 10-engineer team delivering end-to-end digital inspection of 18,000+ assets (deadlegs inspection) with 91% high-quality coverage.
- Actively involved in special projects of organization for UAV, Virtual reality and Underwater robotics ops.

EDUCATION

University of Waterloo

Master of Applied Science (MASC), Mechatronics Engineering (**CGPA: 93.5%**)

Waterloo, Canada

January 2024 – August 2025

- **Thesis link:** “*Intelligent Multi-Robot Autonomy with Connected AMRs and Manipulators for SMART Factory(s)*”, [Transcript](#), [Degree](#)
- **Advisor(s):** [Dr. William Melek](#) and [Dr. George Shaker](#) . Defence committee: [Dr. Soo Jeon](#), and [Dr. Gennaro Notomista](#)

Bahria University

Bachelor of Science (BSc), Electrical (Electronics) Engineering (**CGPA: 80.5%**)

Islamabad, Pakistan

September 2017 – August 2021

- **Thesis link:** “*Path Planning and Perception for multiple mobile robot Swarm Robotics*”, [Transcript](#), [Degree](#)
- **Advisor:** [Dr. Saleem Aslam](#)

SIDE PROJECTS

- **Diffusion Policy vs Imitation learning in Navigation:** [\[Link\]](#) benchmarked diffusion-policy, behavior-cloning, and LSTM imitation learning for TurtleBot 4 navigation; added YOLOv11 for real-time safety detection.
- **Visual Feature Extraction with Machine Learning:** Applied KNN and SVM regression models with FRIQUE image-quality metrics to benchmark visual feature extraction on computer-generated vs. real-world image datasets.
- **Humanoid Robotics: Non-Verbal Human-Robot Interaction** [\[Link\]](#): Designed a non-verbal gesture-recognition model for multimodal control of a NAO humanoid robot assisting in collaborative breakfast preparation.
- **AMRs in Health Care** [\[Link\]](#): Built an RFID-based autonomous nursing robot for pill dispensing, real-time perception, and live video streaming in healthcare environments.
- **UAS Challenge 2022, IMechE UK: Aerial Robotics** [\[Link\]](#): Team Supervisor for a hexacopter UAV (6.9 kg takeoff weight, 100 m operating altitude) for locust-swarm spraying, with a custom carbon-fiber S950 frame and 340 KV BLDC motors delivering 2920 g of thrust.

SELECTED PUBLICATIONS:

- **[C1] M. Qureshi et al.**, “CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception,” *IEEE ICARM 2025*. (**Published**) [\[IEEE Link\]](#)
- **[C2] M. Qureshi et al.**, “Hemispherical Angular Power Mapping of Installed mmWave Radar Modules Under Realistic Deployment Constraints,” *IEEE APS 2026*. (**Published**) [\[arXiv\]](#)
- **[J1] M. Qureshi et al.**, “RAPTAR: Radar Radiation Pattern Acquisition through Automated Collaborative Robotics,” *IEEE Robotics and Automation Letters (RA-L)*. (**submission under revision**) [\[arXiv\]](#)

AWARDS

- **Full Graduate Funding Award**, MASC, University of Waterloo.
- **Top 3 Undergraduate Thesis Project**, Bahria University.
- **Best Performer, Employee of the Month**, Abyss Solutions Inc.
- **3× Hackathon Winner**, Abyss Solutions Robotics & VR Innovation Challenges.

OTHER EXPERIENCE

TA, University of Waterloo

Spring 2024 – Spring 2025

ME 123 (Electrical Eng. for Mechanical Engineers), Spring 2024 | ME 360 (Automatic Control Systems), Fall 2024 & Winter 2025 | MTE (Advanced Calculus), Spring 2025

Conference Presentation, IEEE APS 2026

Jul. 2026

Detroit, Michigan, USA

Conference Presentation, IEEE ICARM 2025

Aug. 2025

Portsmouth, United Kingdom

Symposium Presentation: RoboHub Symposium, Poster & Research Talk

Jan. 2025

University of Waterloo, Canada

Conference Presentation, Rogers “Beyond Connectivity” Summit

Jul. 2024

Brampton, Canada

PEER REVIEW SERVICE

- **Reviewer, IEEE International Conference on Robotics and Automation (ICRA) 2026** (Vienna, Austria; Submission #1761): “*ROR-SLAM: UWB Radar-based Object Recognition for Feature-based SLAM.*”
- **Reviewer, IEEE Sensors Journal** (Manuscript Sensors-69555-2024): “*EAMT-SLAM: Edge-Aided Monocular Thermal Sensor SLAM in Low-Illumination Environments.*”
- **Reviewer, MEIE 2026** (9th Int’l Conference on Mechanical, Electric, and Industrial Engineering; Manuscript MEIE99621): “*Feasibility Analysis and Reinforcement Design of a Portable Radar for Use in Vehicle Environments.*” Recommended minor revisions.